

Mth-643 Final term lect8to15(yasir)

Matrix Algebra: <pre>clear all clc Rowvector=[1,2,3] Columnvector=[1;2;3] threeby3matrix=[1,2,3;3,4,5;7,6,8] transpose(threeby3matrix) %Built in Functions A=pascal(3) %symmetric matrix A^T=A B=magic(3) %antisymmetric F=ones(3) %matrix of ones G=eye(3) %Identity matrix H=zeros(2) % Zero matrix %diagonal matrix from a vector v=[2,3,4]; D=diag(v) E=diag(A) % diagonal entries elehda sy vector mn store %add scalar quantity to each element of vector J=[2,4,5;6,7,1;1,1,1]; K=3+J %Matrix Multiplication 1st n=2nd m syms x y T=[1,2;7,5]; X=[x;y] Z=T*X</pre>	Matrix Algebra: <u>%Matrix Multiplication using for Loop</u> <pre>matrix1=[1;2;3]; matrix2=[1,4,5]; m=3; n=3; for i=1:m for j=1:n c(i,j)=matrix1(i,:)*matrix2(:,j) end end</pre> <u>%Matrix Multiplication Simple method</u> <pre>A=[1;2;3]; %follow multiplication rule B=[1,2,3]; C=A*B</pre> <u>% Multiply Individual Entries</u> <pre>A=[1,2,3];B=[4,3,2]; C=A.*B</pre> <u>%divide Individual entries</u> <pre>A=[2,4,6]; B=[6,8,6]; C=A./B %A oper B nechy D=A.\B %B oper A nechy</pre> <u>%Eigen Values</u> <pre>A=[1,2,3;2,3,5;4,3,2]; eig(A); [V,D]=eig(A)</pre> <u>%Reduced Row Echelon Form</u> <pre>A=[1,2,3;1,3,4;2,5,7]; B=rref(A)</pre>
System of Linear Equations: <pre>%x+2y+4z=1,%3x+2y+4z=2,%2x+4y+5z=5 syms x y z A=[1,2,4;3,2,4;2,4,5]; X=[x;y;z]; B=[1;2;5]; X=inv(A)*B; %1st method X=vpa(A\B); %2nd method X=mldivide(A,B); %3rd method eqn1='x+2*y+4*z=1'; %4th method eqn2='3*x+2*y+4*z=2'; eqn3='2*x+4*y+5*z=5'; sol=solve(eqn1,eqn2,eqn3) X=double([sol.x,sol.y,sol.z])'</pre>	Matrix Equation of Sylvester Type: AX+BX=C <pre>A=[1,2,4;5,6,7;2,3,5]; B=[1,3,4;5,2,3;1,4,6]; C=[1,3,9;3,6,8;2,3,5]; X=sylvester(A,B,C)</pre>
For Loop: Example(1) <pre>%we want to add each element of vector v=1:10; sum=0; for i=1:length(v) sum=sum+i end fprintf('sum of all enteries of vector= %d\n',sum)</pre>	For Loop: Example(2) <pre>%we want to add 1st 30 odd numbers v=1:60; sum=0; for i=1:2:length(v) sum=sum+i end fprintf('sum of 1st 30 odd enteries of vector= %d\n',sum)</pre>
For Loop: Example(3) <pre>%we want to add 1st 30 even numbers v=1:60; sum=0; for i=1:length(v) if(rem(i,2)==0) sum=sum+i; end end fprintf('sum of 1st 30 even enteries of vector= %d\n',sum)</pre>	For Loop: Example(4) <pre>%Calculate factorial num=input('Enter a number= '); fact=1; while num<0 disp('Enter a positive Number') num=input('Enter a number= '); end if num>0 for i=1:num fact=fact*i; end end</pre>

	<pre> end fprintf('The factorial of %d is= %d\n',num,fact) end </pre>
Nested For Loop: Example(5) <pre> %print a pattern rows=input('How many rows do you want to print: '); for R=1:rows for S=1:R fprintf('*') end fprintf('\n') end </pre>	Differentiation and Integration: <pre> %y=x^3 syms x y z f=x^3; df=diff(f,x,2); %differentiation i=int(f,0,2); %Integration %sin(2x) g=sin(2*x); dg=diff(g,x); integ=int(g,x,0,pi/2); %x^2y^3z^2 h=x^2*y^3*z^2; dh=diff(h,z); Ih=int(int(int(h,0,pi),0,pi/2),0,1) </pre>
Solution Of ODES: Example(1) <pre> %dy/dt=ty %y(0)=1 syms y(t) ode=diff(y,t)==t*y; cond=y(0)==1; soly(t)=dsolve(ode,cond) </pre>	Solution Of ODES: Example(2) <pre> %dy/dt+4y(t)=exp(-t) %y(0)=1 syms y(t) ode=diff(y,t)+4*y(t)== exp(-t); cond=y(0)==1; soly(t)=dsolve(ode,cond) </pre>
Solution Of ODES: Example(3) <pre> %d^2y/dt^2=cos(2t)-y %y(0)=1 %y'(0)=0 syms y(t) Dy=diff(y); ode=diff(y,t,2)==cos(2*t)-y; cond1=y(0)==1; cond2=Dy(0)==0; cond=[cond1 cond2]; soly(t)=dsolve(ode,cond) simplsol=simplify(soly(t)) </pre> Download More Files from VUAnswer.com	Solution Of ODES: Example(4) <pre> %d^3u/dx^3=u %u(0)=1 %u'(0)=-1 %u''(0)=pi syms u(x) Du=diff(u); D2u=diff(u,2); ode=diff(u,x,3)==u; cond1=u(0)==1; cond2=Du(0)==-1; cond3=D2u(0)==pi; cond=[cond1 cond2 cond3]; solu(x)=dsolve(ode,cond) simplsolu(x)=simplify(solu(x)) </pre>
System Of ODES: Example(1) <pre> %du/dt=3u+4v %dv/dt=-4u+3v syms u(t) v(t) ode1=diff(u,t)==3*u+4*v; ode2=diff(v,t)==-4*u+3*v; ode=[ode1 ode2]; sol=dsolve(ode); sol.u; sol.v; %OR [usol vsol]=dsolve(ode) </pre>	Ode45:Example(1) <pre> %F=3exp(-t) %tspan=[0,5] %x0=0 %F=3exp(-t) %tspan=[0,5] %y0=0 syms t F=@(t,F) 3*exp(-t); tspan=[0,5]; f0=0; [t,F]=ode45(F,tspan,f0) plot(t,F) </pre>
Ode45:Example(2) <pre> %y'=2t %tspan=[0,5] %y0=0 yprime=@(t,y) 2*t; tspan=[0,5]; y0=0; [t y]=ode45(yprime,tspan,y0) plot(t,y) xlabel('t-axis'); ylabel('Y-axis'); title('Graph of yprime=2t') </pre>	Ode45:Example(3) <pre> %u'=2u+v+5w+exp(-2t) %v'=-3u-2v-8w+2exp(-2t)-cos(3t) %w'=3u+3v+2w+cos(3t) %tspan=[0,5] %u(0)=1 %v(0)=-1 %w(0)=0 clear all clc close all syms u v w t uprime=2*u+v+5*w+exp(-2*t); vprime=-3*u-2*v-8*w+2*exp(-2*t)-cos(3*t); wprime=3*u+3*v+2*w+cos(3*t); tspan=[0,5]; u0=1; v0=-1; w0=0; </pre>

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cond=[u0,v0,w0];
ode=@(t,x)[2*x(1)+x(2)+5*x(3)+exp(-2*t);-
3*x(1)-2*x(2)-8*x(3)+2*exp(-2*t)-
cos(3*t);3*x(1)+3*x(2)+2*x(3)+cos(3*t)]
[t,x]=ode45(ode,tspan,cond)
plot(t,x)
xlabel('X-axis')
ylabel('Y-axis')
title('Plot of Solution Curves')
```

Solution of differential equation using numerical method:

Euler's Method:

$$y(x+h) = y(x) + hy'(x)$$

$$y(x+h) = y(x) + hf(x,y)$$

```
%Euler Method
%y(x+h)=y(x)+hy'(x)
%y(x+h)=y(x)+hf(x,y)
%f(x,y)=x/y
%Euler Method
%y(x+h)=y(x)+hf(x,y)
%f(x,y)=x/y %a=0 %b=2 %y0=1 %x0=0 %n=10
clear all
clc
f=@(x,y)(x/y);
a=0;
b=2;
y0=1;
x0=0;
n=10;
h=(b-a)/n;
x(1)=x0;
y(1)=y0;
for j=1:n
    y(j+1)=y(j)+h*f(x(j),y(j));
    x(j+1)=x(j)+h;
end
plot(x,y)
hold on
syms Y(X)
ode=diff(Y,X)==X/Y;
cond=Y(0)==1;
S=dsolve(ode,cond)
F=matlabFunction(S);
fplot(F,[0,2],'r',2)
hold off
```

Trapezoidal Rule:

$$\int_a^b f(x)dx = \frac{\Delta x}{2} \left[f(x_0) + f(x_n) + 2 \sum_{j=1}^{n-1} f(x_j) \right]$$

```
%Trapezoidal Rule
%find area of curve x^2
%a=0,%b=2, %n=3
f=@(x)x^2;
a=0;
b=2;
n=3;
h=(b-a)/n;
sum=0;
x(1)=1;
for i=1:n-1
    x(i)=a+i*h;
    y(i)=f(x(i));
    sum=sum+y(i);
end
Area=h/2*[f(a)+f(b)+2*sum]
%find area of curve exp(x^2)
f=@(x)exp(x^2);
a=0;
b=2;
n=20;
h=(b-a)/n;
sum=0;
x(1)=1;
for i=1:n-1
    x(i)=a+i*h;
    y(i)=f(x(i));
    sum=sum+y(i);
end
Area=h/2*[f(a)+f(b)+2*sum]
```

Simpson's 1/3 Rule:

$$\int_a^b f(x)dx = \frac{h}{3} \left[f(x_0) + f(x_n) + 2 \sum_{j=1}^{\frac{n}{2}-1} f(x_{2j}) + 4 \sum_{j=1}^{\frac{n}{2}} f(x_{2j-1}) \right]$$

```
%Simpson1by3rule
%y=x^3 %a=0, %b=2, %n=10,
clear all
clc
%
f=@(x)x^3;
a=0;
b=2;
n=10; % n must even
h=(b-a)/n;
if (rem(n,2)~=0)
    fprintf('\n Enter Valid n');
    n=input('\n Enter n as Even');
end
sum1=0;
```

Simpson's 3/8 Rule:

$$\int_a^b f(x)dx = \frac{3h}{8} \left[f(x_0) + f(x_n) + 2 \sum_{j=1}^{\frac{n}{3}-1} f(x_{3j}) + 3 \sum_{i \neq 3k}^{n-1} f(x_i) \right]$$

```
%Simpson 3/8 rule
%exp(2x)sin(6x)
clear all
clc
%f=@(x)exp(2*x).*sin(6*x);
f=@(x)1/(1+x);
a=0;
b=4;
n=12; % n must be multiple of 3
if (rem(n,3)~=0)
```

<pre> sum2=0; for k=1:n-1 x(k)=a+k*h; if (rem(k,2)==0) sum1=sum1+f(x(k)); else sum2=sum2+f(x(k)); end end Area=(h/3)*(f(a)+f(b)+2*sum1+4*sum2); fprintf('Area by Simpson1/3 is=%f',Area) %comparison syms y f1=y^3; Area2=int(f1,a,b) Download More Files from VUAnswer.com Download More Files from VUAnswer.com </pre>	<pre> fprintf('\n Enter valid value of n'); n=input('\n Enter n as multiple of 3: ') end h=(b-a)/n; sum1=0; sum2=0; for k=1:n-1 x(k)=a+k*h; if (rem(k,3)==0) sum1=sum1+f(x(k)); else sum2=sum2+f(x(k)); end end Area=(3*h/8)*(f(a)+f(b)+2*sum1+3*sum2); fprintf('AreabySimpson 3/8 rule = %f',Area); %Comparison syms y f1=1/(1+y); Area2=int(f1,a,b) double(Area2) vpa(Area2,6) </pre>
Newton Raphson Method: $x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$ <pre> %Newton Raphson Method % f(x)=cos(x)-x^3, clear all clc syms x f=@(x)x^3-2*x-5; df=@(x)3*x^2-2; x0=2; tolerance=10^-4; n=5; if df(x0)~=0 for i=1:n x1=x0-f(x0)/df(x0); fprintf('x%d= %f\n',i,x1) if abs(x1-x0)<tolerance break end end end </pre>	Bisection Method: <pre> % x^3-x^2-2 %a=1 and b=2 %f(1)=-2 %f(2)=4 f=@(x)x^3-x-2; a=1; b=2; tol=10^-4; if f(a)*f(b)>0 disp('Bisection method not applicable'); return end disp('Iter No a b c') i=1; while (abs(a-b)>=tol) c=(a+b)/2; if f(c)==0 fprintf('Root of function is %f',c); return end fprintf('%2i \t %f \t %f \t %f\n',i, a, b, c); if f(a)*f(c)>0 a=c; else b=c; end i=i+1; end m=(a+b)/2 fprintf('Root lies at %f/',m) </pre>